

A Multi-task Deep Learning Algorithm for Sleep Stage Scoring and Sleep Arousal Detection

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Abstract

Sleep is crucial for overall health and well-being. Analyzing sleep stages and the frequency of arousal can enhance our understanding of sleep quality and help protect individuals' sleep health. The objective of this study is to explore the development and application of multi-task deep learning model that can simultaneously detect sleep arousal and score sleep stages. We employed a task-incremental approach to develop the multi-task deep learning model. For development and testing, 1069 polysomnography records were incorporated. Additionally, the model's fairness was assessed across various subgroups, including sex, age, and ethnicity. Beyond performance evaluation, we investigated the intermediate features extracted by the model from the raw ECG signal. The multi-task deep learning model achieved a Cohen's κ of 0.68 for sleep stage prediction and an area under the receiver operating characteristic (AUROC) of 0.94 for arousal detection. Additionally, the model showed consistent performance across different subgroups. The analysis of the interpretability of the intermediate features validated their applicability and adaptability to related tasks in sleep analysis. This study demonstrated the capacity of the multi-task deep learning model to score sleep stages and arousal simultaneously with substantial precision using a single-lead ECG. This research offers a promising avenue for advancing incremental learning and multi-task deep learning models in sleep analysis.

Keywords: deep learning; multitask learning; incremental learning; sleep stage scoring; arousal detection

1 Introduction

Sleep constitutes a natural state of reduced consciousness and physical activity, playing a vital role in regulating the body's circadian rhythm and various physiological processes. Inadequate sleep can compromise the immune system and pose significant health risks (1–3). The American Academy of Sleep Medicine (AASM) classifies sleep into distinct stages, which encompass rapid eye movement (REM), non-REM (N1, N2, N3), and wake (4). Sleep staging involves categorizing these stages into epochs, typically lasting 30 seconds, and this is accomplished by manually visually scoring electroencephalographic (EEG) recordings of brain activity, eye movement, and chin muscle activity obtained through polysomnography (PSG). However, PSG is costly and unsuitable for continuous, long-term monitoring outside of a sleep clinic. Consequently, researchers have explored alternative approaches, such as assessing sleep stages through the measurement of heart rate variability (HRV) via the electrocardiogram (ECG), which offers a cost-effective and non-invasive solution (5–7).

Sleep-disordered breathing events, such as sleep apnea, can lead to cortical arousal during sleep, resulting in sleep fragmentation and daytime fatigue. Currently, cortical arousals are identified using electroencephalography (EEG) in polysomnography (PSG) or level 2 home sleep tests (HST). Some previous studies have employed deep learning algorithms to identify sleep arousal using electrocardiography (ECG) signals (8–11), as arousal can affect cardiac activity (12–14). Analyzing the sleep state and the occurrence of arousals can aid in understanding the sleep condition and safeguarding individuals' sleep health.

While many studies have leveraged deep learning approaches for scoring sleep stages and detection arousal using a single-lead ECG, to our knowledge, none have employed a task-incremental approach to develop a multi-task model for these objectives. The aim of this study is to explore the development and application of multi-task deep learning model that can simultaneously detect sleep arousal and score sleep stages. Employing a task-incremental approach allows for the seamless integration of additional tasks, obviating the need to retrain the foundational model. Utilizing a single-lead ECG simplifies data acquisition and offers potential for seamless integration into home-based health management systems. Additionally, we evaluate the model's fairness by analyzing its performance across different sex, age, and ethnicities. Finally, we investigate the correlation between the model's intermediate features and manually extracted cardiac features to gain insights into the information learned by the model, ensuring the derived features can be applied to further related tasks in sleep analysis.

2 Method

2.1 Data Source and Subject Exclusion

We employed the Multi-Ethnic Study of Atherosclerosis (MESA) dataset for the development, testing, and statistical analyses of our deep learning models. MESA represents a collaborative longitudinal study conducted across multiple centers, aimed at tracking the progression from subclinical to clinical cardiovascular disease (15). The Sleep Exam substudy, conducted between 2010 and 2012, included 2237 participants from the original cohort of 6814 individuals. This diverse group encompassed men and women from various ethnic backgrounds, including Black, White, Hispanic, and Chinese-American populations. The Sleep Exam protocol involved unattended full overnight polysomnography (PSG), 7-day wrist-worn actigraphy, and completion of a comprehensive sleep questionnaire. In our analysis, we utilized the single-lead electrocardiogram (ECG) extracted from the PSG recordings, which were sampled at a frequency

of 256 Hz. The PSG and demographic data of participants can be obtained from <https://sleepdata.org/datasets/mesa>. The MESA dataset was fully anonymized before being accessed. The University of Arizona Institutional Review Board has classified this project as “Not Human Subjects Research” and thus is exempt.

For the detection of cortical arousal, a key focus of our study, certified scorers meticulously applied the American Academy of Sleep Medicine (AASM) criteria (4). These criteria define cortical arousal as a sudden shift in EEG frequency lasting a minimum of 3 seconds, occurring after at least 10 consecutive seconds of sleep. Such shifts could encompass alpha, theta, and/or delta waves, along with frequencies surpassing 16 Hz. Notably, during rapid-eye-movement (REM) sleep, an elevation in the electromyogram (EMG) signal is also stipulated as a requirement.

From the MESA dataset, 2,055 out of the 2,237 subjects possessed PSG records; however, 986 among them were excluded due to unreliable or missing arousal/sleep stage events. Consequently, a cohort of 1,069 subjects remained for our detailed model development and subsequent statistical analyses, outlined in Fig 1. For the construction of the testing set, 20% (214 subjects) was randomly sampled from the 1,069 subjects. The remaining 684 subjects constituted the training set, while a subset of 171 subjects was reserved for validation during the model training phase.

2.2 Data Preprocessing

To streamline the preprocessing phase, we opted for the application of a robust normalizer exclusively intended for the normalization of the ECG signal. The sleep stages and arousal labels were extracted from the XML files and synchronized with the 256 Hz normalized ECG signal. Our labeling approach employed a binary system where 1 indicated the presence of arousal. Following established conventions (5,6), the wake status was denoted as 0. To capture light sleep, we amalgamated the N1 and N2 sleep stages under the unified label 1. Stage N3 was designated as deep sleep, represented by label 3, while REM sleep received label 4.

2.3 Model Architecture and Model Training

Our model was constructed following the DeepCAD methodology employed in a prior investigation for arousal detection via a single-lead ECG signal (11). The model's schematic representation can be found in Figure 2. This architecture translates ECG signals into probabilities assigned to the sleep stage and sleep arousal classes at both the epoch and second levels, respectively.

The architecture of the proposed model consists of two primary components. The initial element is the original DeepCAD architecture, encompassing an inception block, two pre-activation residual blocks, a batch normalization layer, long short-term memory (LSTM) units, and fully connected layers. For arousal detection, the input ECG signal, originally sampled at a frequency of 256 Hz, undergoes downsampling to 1 Hz.

In consideration of the typical 30-second intervals for sleep stage scoring, additional downsampling is essential for accurate sleep stage assessment. In this study, a subnetwork was integrated into the original DeepCAD architecture for sleep stage scoring. This proposed subnetwork leverages the intermediate output from the DeepCAD architecture, generated immediately before the final batch normalization layer. This output is then subjected to additional downsampling, aggregating it into 30-second epochs. Subsequently, a softmax layer is applied to classify the output into distinct sleep stages. This approach enables the sharing of mutual information between the two tasks.

We employed a task-incremental approach to train the model that encompassed two distinct stages. During the initial stage, the model underwent training to predict arousal employing the back-propagation-through-time (BPTT) algorithm coupled with the Adam optimizer. To bolster the model's resistance to input noise, we integrated L2 weight decay alongside a dropout rate of 0.2, serving as regularization techniques. Additionally, batch normalization was implemented to counteract gradient vanishing and enhance overall performance. The calculation of loss relied on the utilization of cross-entropy, as detailed in Equation 1.

$$\mathcal{L}_{Arousal} = -\frac{1}{N} \sum_{i=1}^N y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i) \quad (1)$$

where N signifies the total number of seconds per batch, where y_i represents the binary label, and p_i stands for the predicted arousal probability for the current second. Throughout each training epoch, the validation set was employed to assess the model's performance based on validation loss. Moreover, we introduced a dynamic learning rate strategy, reducing the rate by a factor of 0.1 when validation loss remained steady for four consecutive epochs. The best performing model was trained with a learning rate of $1e-4$, a weight decay rate of 0.1, a dropout rate of 0.2, and the previously outlined architecture. Subsequently, we locked all hyperparameters of the layers responsible for arousal prediction and transitioned to sleep stage scoring. In this phase, we adhered to the same training process as elucidated in the initial step, with the sole alteration being the utilization of the multi-class cross-entropy loss function. The formulation of the multi-class cross-entropy loss function can be represented as Equation 2, where N is the total number of epochs per batch, while C represents the four distinct sleep stages. $y_{i,c}$ designates the ground truth label as c for the sample epoch i , and $p_{i,c}$ corresponds to the predicted probability for the i^{th} epoch being stage i . During the initial training phase for sleep stage scoring, an initial learning rate of 0.01 was utilized. Furthermore, we adopted the Ax library (Meta Platforms, Inc., USA) for the refinement of hyperparameters, aiming to enhance the overall performance of the model.

$$\mathcal{L}_{Sleep-stage} = -\frac{1}{C} \sum_{i=1}^N y_i \cdot \log(p_{i,c}) \quad (2)$$

2.4 Model Performance Evaluation

The neural network employed in this investigation predicted arousal on a per-second basis. Additionally, every 30-second epoch generated four values, representing the probabilities associated with the four sleep stages: wake, light, deep, and REM. To gauge the effectiveness of arousal detection, performance assessment relied on the area under the receiver operating characteristic (AUROC) and areas under precision-recall curves (AUPRC) metrics, while the accuracy and Cohen's κ metric were utilized to appraise sleep stage scoring proficiency. Cohen's κ metric is expressed in Equation 3, where t_p represents the number of correctly predicted epochs, while total signifies the overall count of epochs. Here, k refers to the k -th sleep stage, p_k denotes the count of epochs predicted as the k -th stage, and t_k indicates the count of epochs that are the k -th stage. Cohen's κ serves as a statistic employed to assess inter-rater agreement concerning categorical variables. It contrasts observed agreement with anticipated agreement, accounting for the possibility of chance agreement. This metric ranges from -1 to 1, where a value of 1 indicates perfect agreement and 0 indicates agreement no better than chance. The annotations provided by experts within the MESA dataset were utilized as the gold standard.

$$\kappa = \frac{tp \cdot total - \sum_{k=1}^4 p_k \times t_k}{total^2 - \sum_{k=1}^4 p_k \times t_k} \quad (3)$$

2.5 Statistical Analysis

Clinical measure covariate analyses were executed to assess the variations in performance for both arousal prediction (AUROC and AUPRC) and sleep staging (light, deep, and REM fraction, along with Cohen's κ) in relation to three covariates: sex, age, and ethnicity. For age

categorization, the 25th, 50th, and 75th quantiles served as thresholds, yielding age cutoffs of 61, 66, and 75 years, respectively. Binary covariate (sex) analysis was conducted through separate t-tests, while polychotomous covariates (age and ethnicity) were evaluated using Kruskal-Wallis tests. We considered that p-value <0.05 indicated statistical significance in analysis.

An exploration was also conducted into the features extracted by the model from the raw ECG signal. The calculation of the Spearman correlation coefficient served to assess the relationship between the heart rate information derived from the ECG signals and the intermediate output of DeepCAD. This intermediary output was applied in both the sleep stage scoring and arousal detection tasks. The Spearman correlation coefficient, functioning as a gauge of the magnitude and direction of the association between two variables, is expressed in Equation 4, where σ_X is the standard deviation of X, σ_Y is the standard deviation of Y, $cov(X, Y)$ denotes the covariance of the X and Y.

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sigma_X \sigma_Y} \quad (4)$$

The Spearman coefficient is a distribution-free test, unburdened by assumptions about the sample data's distribution. As such, it proves fitting for scenarios where one variable is numeric ordinal, and the other is continuous, or when both variables are continuous (16).

To investigate the intermediate features extracted by the model from the raw ECG signal, the initial step involved segmenting the ECG signals into 30-second windows. The Pan-Tompkins algorithm was employed to extract the R-peaks from these windows. Subsequently, we computed the instantaneous heart rate, alongside time and frequency domain HRV features like SDNN, RMSSD, and pNN50. Ultimately, a Spearman correlation analysis was performed, assessing the relationships between each of the 288 channels of the intermediate output and the derived features.

3 Results

3.1 Subject Characteristics

Table 1 provides an overview of the subjects' characteristics, encompassing arousal duration extracted from annotation files. Across the training, testing, and validation sets, the subjects were similar in terms of age, AHI, sleep duration, and the distribution of sleep stages. Nevertheless, there were disparities in the distributions of sex and ethnicity.

3.2 Model Performance

The output of the deep learning model for arousal prediction and sleep stages is depicted in Figure 3 and Figure 4, respectively. The arousal output presented in Figure 3 corresponds to a 90-second segment. In Figure 3, the upper panel showcases the input ECG signal, the middle panel depicts the binary ground truth indicating arousal occurrences and the lower panel displays the model's probability predictions at a 1-second resolution. As sleep are being staged every 30 seconds, Figure 4 demonstrates the ground truth (above) and predicted sleep stage (below) of the hypnogram for 1.5 hours.

Using the held-out test set from the MESA dataset comprising 214 subjects, the model achieved an AUROC of 0.94 and AUPRC of 0.58 for arousal identification. In terms of sleep stage scoring, the overall accuracy across the four classes reached 0.79, with a Cohen's κ value of 0.68. The model's confusion matrices for the testing set are showcased in Figure 5. Specifically, the algorithm demonstrated per-class performance of (recall, precision) as follows: wake-(0.85, 0.99), light-(0.85, 0.76), deep-(0.28, 0.50), and REM-(0.78, 0.72) on the testing set. This indicated a

proficient capability of the model in accurately identifying and classifying the wake, light, and REM stages.

A summary of the statistical analysis results for arousal prediction metrics among various subgroups is presented in Table 2. The testing set comprises 105 male and 109 female subjects, as indicated in the same table. For female subjects, the AUROC and AUPRC values for arousal detection were 0.93 ± 0.04 and 0.55 ± 0.13 , while for male subjects, these values were 0.93 ± 0.03 and 0.57 ± 0.15 , respectively. A t-test was conducted, revealing no significant differences between male and female subjects concerning arousal AUROC (p-value = 0.49) or AUPRC (p-value = 0.21).

The age subgroups were defined as follows: (a) ≤ 60 , (b) 61 – 75, (c) 66 – 75, and (d) > 75 , encompassing 44, 48, 70, and 52 subjects, respectively. Kruskal-Wallis tests were performed, showing no significant differences in terms of AUROC (p-value = 0.14) and AUPRC (p-value = 0.08) across the four age subgroups.

In terms of ethnic distribution, the sample encompassed 81 White/Caucasian, 25 Chinese American, 52 Black/African American, and 26 Hispanic participants. Among these subpopulations, White/Caucasian participants exhibited the highest performance, achieving an AUROC of 0.94 ± 0.03 and an AUPRC of 0.56 ± 0.13 for arousal detection. Chinese American and Black/African American participants demonstrated comparable performance, with Chinese Americans achieving an AUROC of 0.93 ± 0.05 and an AUPRC of 0.54 ± 0.17 , and Black/African American achieving an AUROC of 0.927 ± 0.037 and an AUPRC of 0.54 ± 0.14 for arousal detection. Hispanic participants achieved an AUROC of 0.93 ± 0.03 and an AUPRC of 0.57 ± 0.13 for arousal detection. The estimated distributions of AUROC and AUPRC for arousal prediction are depicted in Figure 6. Using Kruskal-Wallis tests, no statistically significant differences were observed in arousal AUROC (p-value = 0.51) and arousal AUPRC (p-value = 0.72).

The performance of sleep stage scoring was assessed through the computation of Cohen's κ values, allowing for comparisons across diverse subgroups, as summarized in Table 3. Notably, no significant differences emerged within the two-sex, four-age, and four-ethnicity cohorts. Specifically, the mean Cohen's κ of female subjects exhibited a slightly higher value than that of male subjects (p-value = 0.34). Concerning the age cohorts, Cohen's κ values declined with age, as evidenced by the highest mean value of 0.62 observed in the ≤ 60 cohort, while the > 75 cohort registered the lowest mean value of 0.57. However, the Kruskal-Wallis test revealed no statistically significant variations in these mean values (p-value = 0.33). Turning to the 4 ethnicity cohorts, the Hispanic subgroup demonstrated the highest mean Cohen's κ of 0.63, similar to the White cohort's similar mean Cohen's κ of 0.61. In comparison, the Black and Chinese American cohorts attained mean Cohen's κ values of 0.57 and 0.58, respectively. Nevertheless, no substantial differences emerged between the four ethnicity groups (p-value = 0.20).

In order to scrutinize the correlations between our model predictions and expert-scored PSG data, we conducted a comparison of sleep stage fractions (see Figure 7). Generally, our model exhibited good concordance with the PSG data, although there was a minor tendency for underestimating deep sleep in females (Figure 7 (A) - (C)). Furthermore, our observations revealed a pattern in which males displayed a higher proportion of light sleep (N1/N2) and a relatively lower proportion of deep sleep (N3/N4) and REM, a trend that resonated with our predictions.

Regarding ethnicity correlations, we noted an overestimation of the light sleep fraction among Chinese Americans and an underestimation of deep sleep for both White and Chinese American subpopulations (Figure 7 (D) and (E)). However, the relative effects on other sleep stages and ethnicities were comparable to those observed in the reference sleep stages.

When examining age correlations, we found that our model exhibited significant and comparable effects on predicting light sleep, deep sleep, and REM, similar to its impact on reference sleep stages. Nonetheless, the absolute values were slightly inflated, and there was a minor propensity for overestimating light sleep in individuals aged over 75 (Figure 7 (G) - (I)).

3.3 Correlation Analyses Between Intermediate Outputs and Cardiac Signals

In order to derive the instantaneous heart rate and HRV features from the ECG signal, we employed the Pan-Tompkins algorithm (17). This algorithm encompasses a series of procedural steps, including a Butterworth bandpass filter (with low and high cutoff frequencies set at 8 Hz and 20 Hz, respectively), signal differentiation, and moving window integration. These steps collectively facilitate the identification of R peaks within the ECG signal.

We explored the correlation between manually extracted features (instantaneous heart rate, SDNN, RMSSD, and pNN50) and the outputs of all 288 channels in our model. Using a correlation coefficient threshold of 0.5, we identified channels with an absolute Spearman correlation coefficient exceeding the set threshold. Among the various features, instantaneous heart rate demonstrated a robust correlation with a part of the intermediate channel outputs. As illustrated in Figure 8, four channels (115, 133, 177, and 218) effectively captured heart rate variations in over 175 subjects (more than 80%) within the testing set, while 17 channels exhibited a significant correlation with instantaneous heart rate in at least 25% of the subjects.

We computed the mean Spearman correlation coefficient between the output of each intermediate channel and the instantaneous heart rate for each subject, illustrating the overall signal relationship. To achieve this, we calculated 214 coefficients per channel and generated a histogram of the outcomes, visually represented in Figure 9. The histogram predominantly displayed average coefficients below 0.5, suggesting a limited correlation between the intermediate outputs and instantaneous heart rate. Nonetheless, we observed that four channels exhibited average coefficients surpassing 0.6, signifying a substantial correlation with heart rate.

We investigated the correlation between the intermediate channel output and other extracted time domain HRV features. While numerous channels displayed correlations with manually extracted features from certain subjects, the number of subjects showing such correlations was notably fewer than those related to instantaneous heart rate. Hence, our focus centered on reporting the channels that captured HRV feature variations in at least 21 subjects (equivalent to 10% of the testing subjects). The outcomes are summarized in Table 4.

We present the intermediate channels that exhibited notable correlations with manually extracted HRV features in Table 4. Importantly, a significant portion of these channels also displayed robust correlations with instantaneous heart rate, as illustrated in Figure 8. Specifically, channels 115, 133, and 169 stood out for demonstrating particularly strong correlations with both HRV features and instantaneous heart rate. Remarkably, channels 115 and 133 displayed correlations with multiple HRV features in some subjects. Figure 10 displays the intermediate channel outputs of 133 in relation to the instantaneous heart rate of subject 117. Meanwhile, Figure 11 illustrates the intermediate channel outputs of 115 in relation to the SDNN feature of subject 80. However, it's essential to highlight that our reporting only encompasses channels capturing variations in no fewer than 21 subjects, equating to 10% of the testing set. Notably, channels 222 and 253 exhibited more robust correlations with RMSSD and SDNN, respectively, when compared to other HRV features or the instantaneous heart rate.

4 Discussion

In this paper, we developed a deep learning model capable of simultaneously scoring sleep stages and identifying sleep arousals utilizing a single-lead ECG. Built upon the foundation laid by a previous investigation (11), our proposed multi-task deep learning model, trained on the PSG records of 684 subjects and test on 214 subjects, replicated an AUROC of 0.94 and an AUPRC of 0.58 for sleep arousal detection. Additionally, it achieved an average accuracy of 0.79 with a Cohen's κ of 0.68 for 4-label sleep stage scoring. We also thoroughly examined the model's impartiality across diverse cohorts by scrutinizing the predictive performance for arousal among various factors such as sex, age, and ethnicity. The outcomes revealed no significant variance in the model's performance across these cohorts. Moreover, benchmarked against expert-scored PSG as the gold standard, the sleep staging model consistently demonstrated proficient performance across varying age groups, genders, and ethnicities. Delving into model interpretability, we explored the correlation between the intermediate channel output and several manually extracted features encompassing instantaneous heart rate, SDNN, RMSSD, and pNN50. While certain channels exhibited correlations with SDNN, RMSSD, and pNN50, the intermediate output channel displayed a markedly stronger correlation with instantaneous heart rate compared to any other feature investigated in this study.

Numerous prior investigations have concentrated on either sleep arousal detection or sleep stage scoring. Deep learning models for sleep arousal detection have commonly utilized PSG data as input (8,18–20). Notable examples include the work by Liu et al. (10) and Howe-Patterson et al. (19), achieving an AUROC of 0.95 and AUPRC of 0.57, respectively. Some researchers have employed feature engineering alongside secondary-analyzed signals to enhance performance. For instance, Olsen et al. (8) leveraged HRV features and sleep stages, attaining a precision of 0.72 and sensitivity of 0.63. Basner et al. (9) initially extracted RR intervals and achieved a sensitivity of 0.69. In our previous study, we also proposed an end-to-end deep learning model that utilized a single-lead ECG to detect arousal, achieving an AUPRC of 0.62 and an AUROC of 0.93 on the MESA dataset (11). Regarding sleep staging, Radha et al. (21) employed ECG training and subsequent transfer learning to PPG, achieving an overall accuracy of 0.77 and a κ of 0.65. Fonseca et al. (6) reported an overall accuracy of 0.76 and a κ of 0.6 utilizing ECG input. Some studies ventured into advanced ECG-based feature engineering to enhance performance. Sridhar et al. (5) integrated instantaneous heart rate with a diluted CNN for the 4-label sleep stage scoring. The model, trained on sleep heart health study and MESA datasets, achieved an accuracy of 0.80 and a κ of 0.69 on the MESA dataset. Radha et al. (7) achieved a similar performance—0.77 for overall accuracy and 0.61 for κ —by using HRVs extracted from ECG. In contrast, our proposed multi-task model, trained on the PSG records of 684 subjects, stands out by leveraging shared representations to improve data efficiency and mitigate model overfitting, delivering performance that is either superior or comparable to the aforementioned single-task models. Additionally, the model has the ability to be expanded to accommodate more tasks using the task-incremental strategy, obviating the need to retrain the foundational model. Furthermore, by utilizing input from only a single-lead ECG, our model holds substantial promise for seamless integration into home-based health management systems.

We evaluated our model's performance across various cohorts defined by sex, age, and ethnicity, recognizing the significance of equity in the development of health management and biomedical decision systems. Our assessment encompassed comparisons of AUROC and AUPRC for arousal prediction and Cohen's κ for sleep staging. Despite discernible numerical discrepancies, statistical analyses revealed no significant performance disparities among the categories of sex, age, or ethnicity. Notably, previous research has established that males exhibit notably higher proportions of light sleep but reduced proportions of deep sleep and REM, with

deep sleep and REM declining with age (22). Impressively, our model accurately emulated these anticipated trends within the predicted stages. However, within the testing set, subjects above the age of 75 manifested a decline in the fraction of light sleep. This observation could possibly be attributed to diminishing sleep quality, as evidenced by reductions in all sleep stages—light, deep, and REM. This, in turn, indicates an elevation in the wake fraction with increasing age. Another plausible factor influencing the less-than-optimal prediction of deep sleep could be the limited number of deep sleep labels available.

The statistical analysis of shared intermediate outputs across both classification tasks revealed the ability of our deep learning model to extract bothprecedented and unprecedented features from ECG signals. Overall, 17 channels exhibited correlations with instantaneous heart rates in at least 25% of the subjects. Notably, four specific channels—115, 133, 169, and 218—demonstrated robust correlations with the instantaneous heart rates of at least 80% of the subjects. Given the well-established association between heart rate, sleep stage transitions, and arousal occurrences, our findings underscore the pivotal role of instantaneous heart rate in identifying changes in sleep stages or the onset of arousal (5,7,11). We extended our investigation to encompass more intricate HRV features, including SDNN, RMSSD, and pNN50. While a smaller subset of subjects' HRV features was captured by the model, it's important to note that channels exhibiting strong correlations with instantaneous heart rates also displayed correlations with these HRV features. The SDNN, RMSSD, and pNN50 are derived from RR intervals and are inversely related to instantaneous heart rates, likely explaining our observations. Nonetheless, some channels displayed correlations solely with HRV features, such as 222 with RMSSD and 253 with SDNN. Notably, the additional processing of RR intervals renders HRV features less noisy compared to instantaneous heart rates, as illustrated in Figure 11. Lastly, it's worth acknowledging that the employment of a sliding window in our analysis may have introduced a slight delay between the intermediate channel output and calculated HRV features, possibly contributing to a reduction in the correlation coefficient. Through this extensive analysis and investigation of the intermediate features and their correlations, we have elucidated the interpretability of the features extracted by our model, affirming their potential applicability and adaptability to a variety of related tasks in sleep analysis and beyond.

Our study offers significant insights into the application of deep learning for sleep staging, yet several limitations warrant consideration. Firstly, our conclusions draw exclusively from the MESA dataset's testing set, thereby constraining the generalizability of our findings. To establish robustness, further examinations across diverse datasets are imperative. Secondly, the accuracy of predicting deep sleep stages using our model notably trailed that of other stages. A plausible explanation for this disparity is the relatively lower inter-rater reliability among experts for deep sleep, known to exhibit substantial variability with age and sex. Lastly, our utilization of a 30-second sliding window for feature computation may have restricted the range of acquired features, prompting the need for an extended window size to encompass additional frequency domain HRV features. However, this alteration could lead to reduced correlation coefficients due to diminished prediction resolution.

5 Conclusion

In conclusion, our study presents a deep learning model's potential to simultaneously address sleep staging and arousal identification using ECG signals. The model showcases robust performance, proving its capacity to accurately predict sleep stages and arousal. Moreover, it demonstrates fairness across various demographic subgroups, essential for equitable health

management systems. By uncovering automatic feature extraction capabilities, our findings emphasize the model's interpretability, marked by its ability to correlate with key physiological indicators like instantaneous heart rate and heart rate variability. This interpretability elucidates the model's mechanisms, confirming the applicability and adaptability of the extracted intermediate features to related tasks in sleep analysis. Although certain limitations, such as dataset exclusivity and optimal feature window size, were acknowledged, our study propels the integration of deep learning into comprehensive sleep analysis frameworks. This research contributes valuable insights to the field and presents a promising avenue for advancing incremental learning and multi-task deep learning model in sleep analysis.

Acknowledgement

This work was supported by grants from the National Heart, Lung, and Blood Institute (\#R21HL159661), and the National Science Foundation (\#2052528)

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Figure Captions

Figure 1. Flow chart of data inclusion in this study. REM: rapid eye movement; NREM: non-rapid eye movement.

Figure 2. The full architecture of the deep learning model for sleep stage scoring and arousal detection. AvePool: Average Pooling; BN: Batch Normalization; Conv1D: 1d Convolution; FC: Fully Connected layer; LSTM: Long Short-Term Memory; ReLU: Rectified of Linear Unit.

Figure 3. Illustrative arousal output of a 90-second segment.

Figure 4. Example of labels and predictions of sleep stage scoring of a 1.5h segment.

Figure 5. Confusion matrix of sleep stage scoring.

Figure 6. Arousal area under receiver operating characteristic (AUROC) and areas under precision-recall curves (AUPRC) comparison based on different subpopulations.

Figure 7. Sleep fraction comparison between model prediction and ground truth.

Figure 8. The number of subjects that have a correlation coefficient >0.5 between instantaneous heart rate intermediate channel output versus channel.

Figure 9. Histogram of intermediate output channel versus the Spearman correlation coefficient between channel output and instantaneous heart rate.

Figure 10. Sample intermediate output channel [133] and instantaneous heart rate of subject [117] for a segment of 200s Sample intermediate output channel [115] and SDNN of subject [80] for a segment of 100s.

Figure 11. Sample intermediate output channel [115] and SDNN of subject [80] for a segment of 100s.

Table 1: Descriptive characteristics of the subjects

Characteristics	Training (n = 684)	Validation (n = 171)	Testing (n = 214)
Female (%)	54.09%	50.93%	54.39%
AHI (with $\geq 3\%$ oxygen desaturation or with arousal) (mean $\pm$ SD)	22.94 $\pm$ 19.58	22.88 $\pm$ 18.54	22.83 $\pm$ 17.76
Age (mean $\pm$ SD)	69.01 $\pm$ 8.84	69.06 $\pm$ 9.35	68.67 $\pm$ 8.68
Race			
Black, African American	26.02%	26.32%	24.30%
Caucasian, White	39.77%	45.03%	37.85%
Chinese American	11.70%	9.36%	11.68%
Hispanic	22.51%	19.30%	26.71%
Arousal index (mean $\pm$ SD)	22.08 $\pm$ 12.01	21.67 $\pm$ 10.70	23.16 $\pm$ 11.58
Sleep duration [hour] (mean $\pm$ SD)	6.11 $\pm$ 1.30	6.16 $\pm$ 1.28	6.07 $\pm$ 1.22
Awake stage percentage (%)	20.68%	20.49%	21.54%
Light sleep stage percentage (%)	55.74%	55.61%	55.19%
Deep sleep stage percentage (%)	8.69%	8.33%	8.87%
REM stage percentage (%)	14.89%	15.57%	14.40%

AHI: Apnea-Hypopnea Index; SD: Standard Deviation; REM: Rapid Eye Movement.

Table 2: Summary of arousal prediction metrics between different subgroups

Covariates	Subgroup	Count	Metric	Mean (SD)	p- value
Sex	Female	109	AUROC	0.93±0.04	0.49
	Male	105		0.93±0.03	
	Female	109	AUPRC	0.55±0.13	0.21
	Male	105		0.57±0.15	
Age	<= 60	44	AUROC	0.94±0.03	0.14
	61–65	48		0.94±0.04	
	66–75	70		0.93±0.03	
	> 75	52		0.93±0.04	
	<= 60	44	AUPRC	0.56±0.12	0.08
	61–65	48		0.60±0.14	
	66–75	70		0.53±0.13	
	> 75	52		0.56±0.16	
Ethnicity	Black	52	AUROC	0.93±0.04	0.51
	Chinese American	25		0.93±0.05	
	Hispanic	56		0.93±0.03	
	White	81		0.94±0.03	
	Black	52	AUPRC	0.54±0.14	0.72
	Chinese American	25		0.54±0.17	
	Hispanic	56		0.57±0.13	
	White	81		0.56±0.13	

AUROC: Area Under the Receiver Operating Characteristic; AUPRC: Area Under Precision-Recall Curve; SD: Standard Deviation;

Table 3: Comparison of Cohen's κ between different subgroups

Covariates	Subgroup	Count	Mean(SD)	p- value
Sex	Female	109	0.61±0.16	0.34
	Male	105	0.59±0.16	
Age	<= 60	44	0.62±0.17	0.33
	61–65	48	0.62±0.14	
	66–75	70	0.60±0.16	
	> 75	52	0.57±0.16	
Ethnicity	Black	52	0.57±0.18	0.20
	Chinese American	25	0.58±0.15	
	Hispanic	56	0.63±0.16	
	White	81	0.61±0.15	

SD: Standard Deviation

Table 4: Count of subjects that showed a Spearman correlation coefficient > 0.5 between intermediate output channel regarding HRV features.

Intermediate Channel #	HRV Feature	Subjects Count
40	RMSSD	23
88	RMSSD	24
108	RMSSD	38
	pNN50	22
115	RMSSD	54
	pNN50	30
133	RMSSD	28
	pNN50	22
169	RMSSD	30
	pNN50	22
177	RMSSD	25
222	RMSSD	32
253	SDNN	38

HRV: Heart Rate Variability; pNN50: Number of successive RR intervals pairs differing more than 50 ms divided by the total number of RR intervals; RMSSD: Square root of the mean squared differences between successive RR intervals; SDNN: Standard deviation of normal RR intervals.

Figure 1

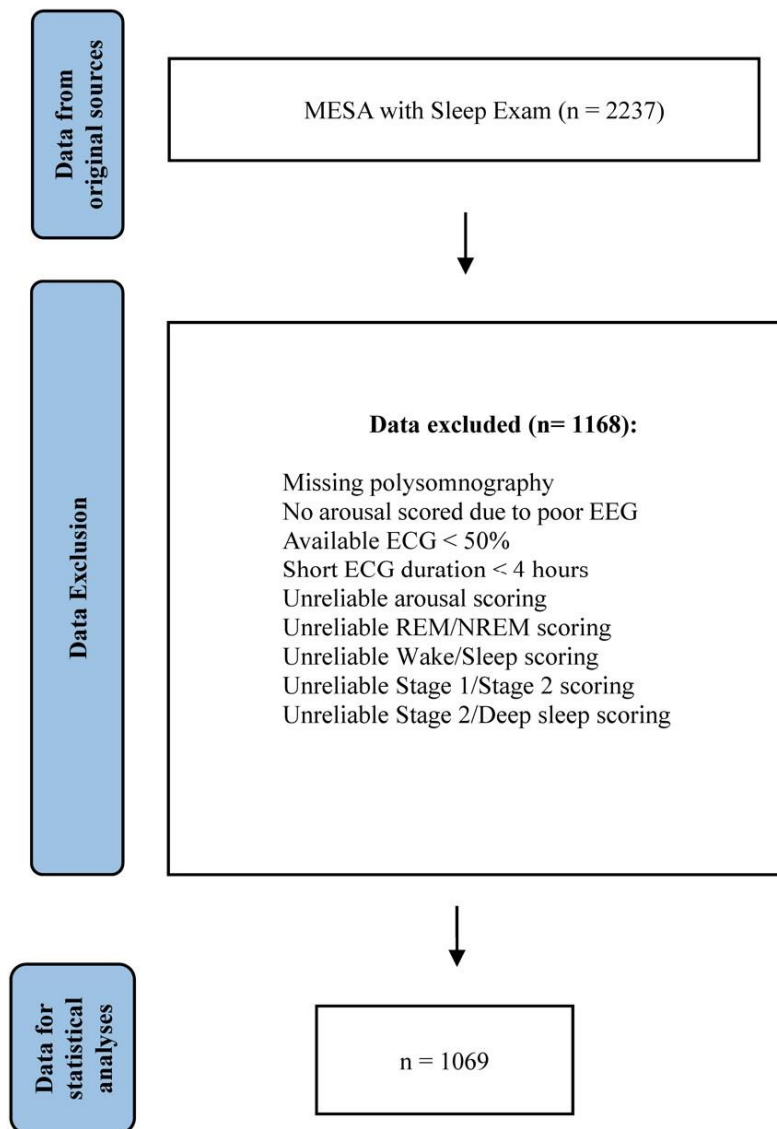


Figure 2

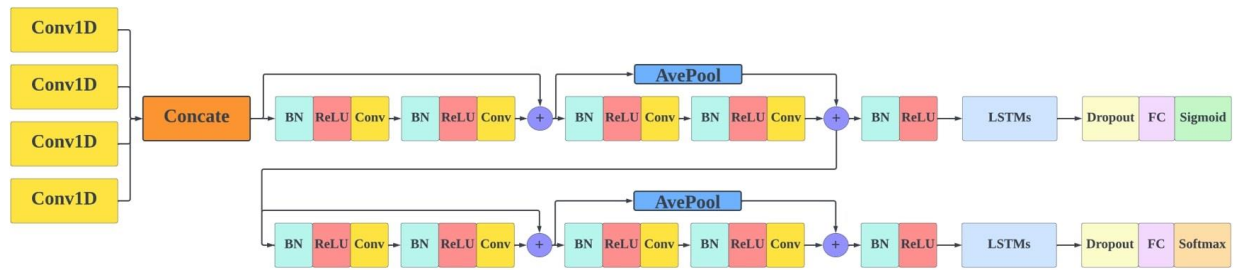


Figure 3

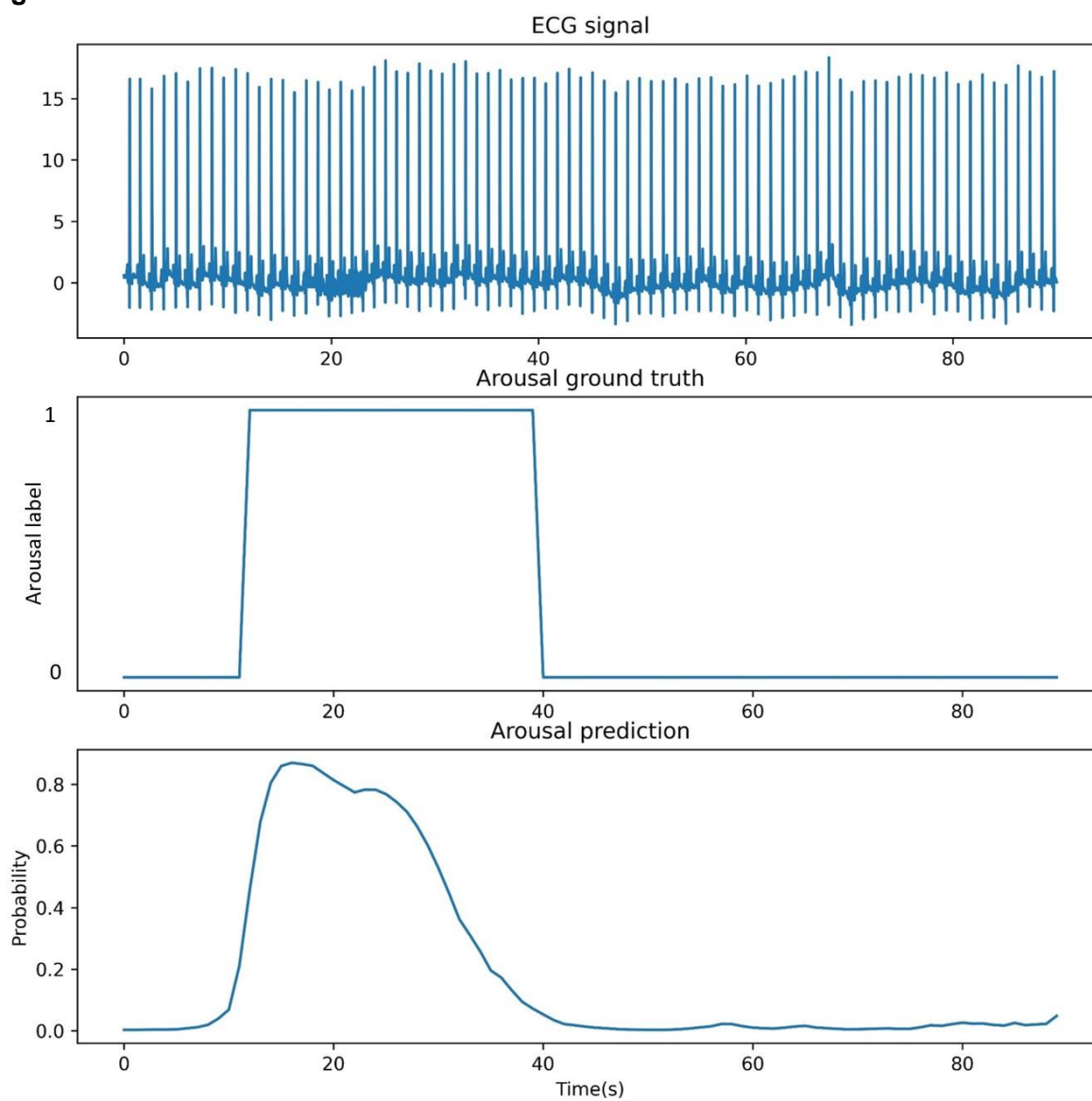


Figure 4

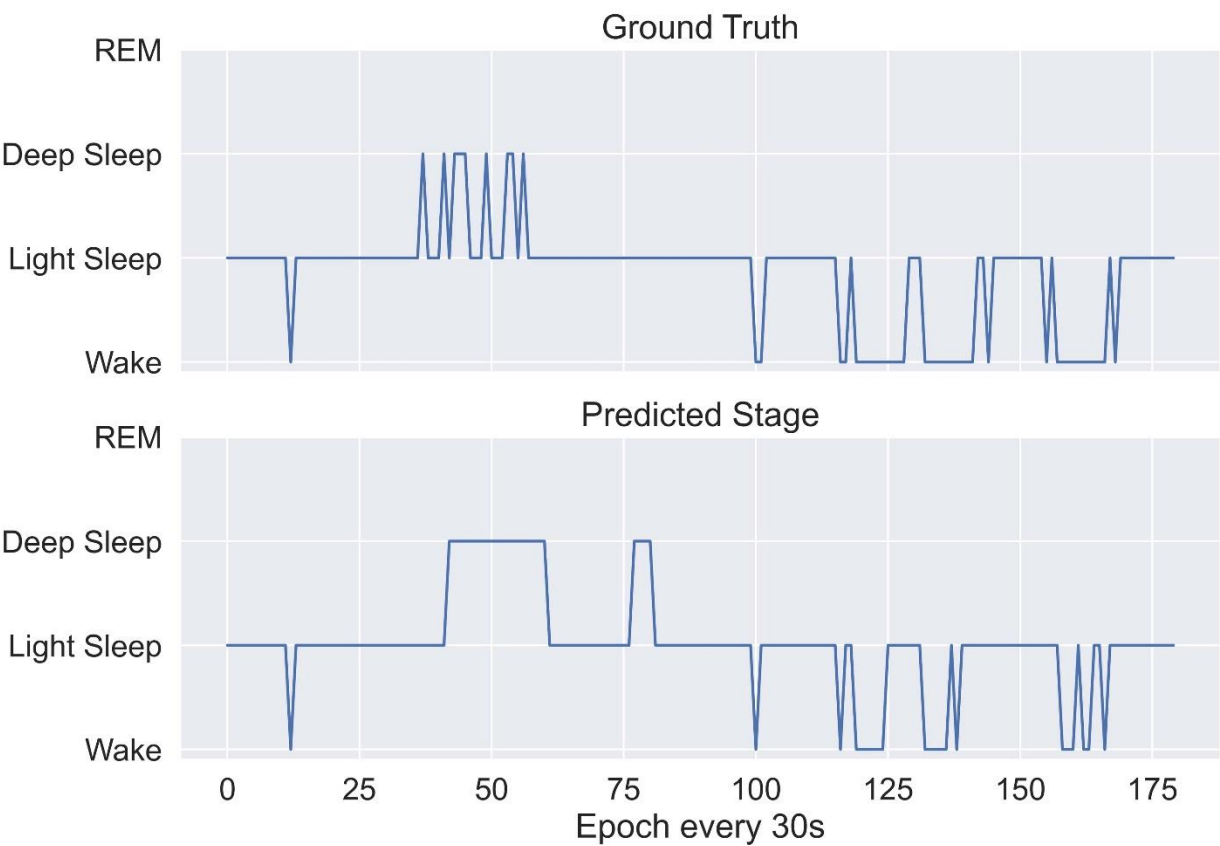


Figure 5

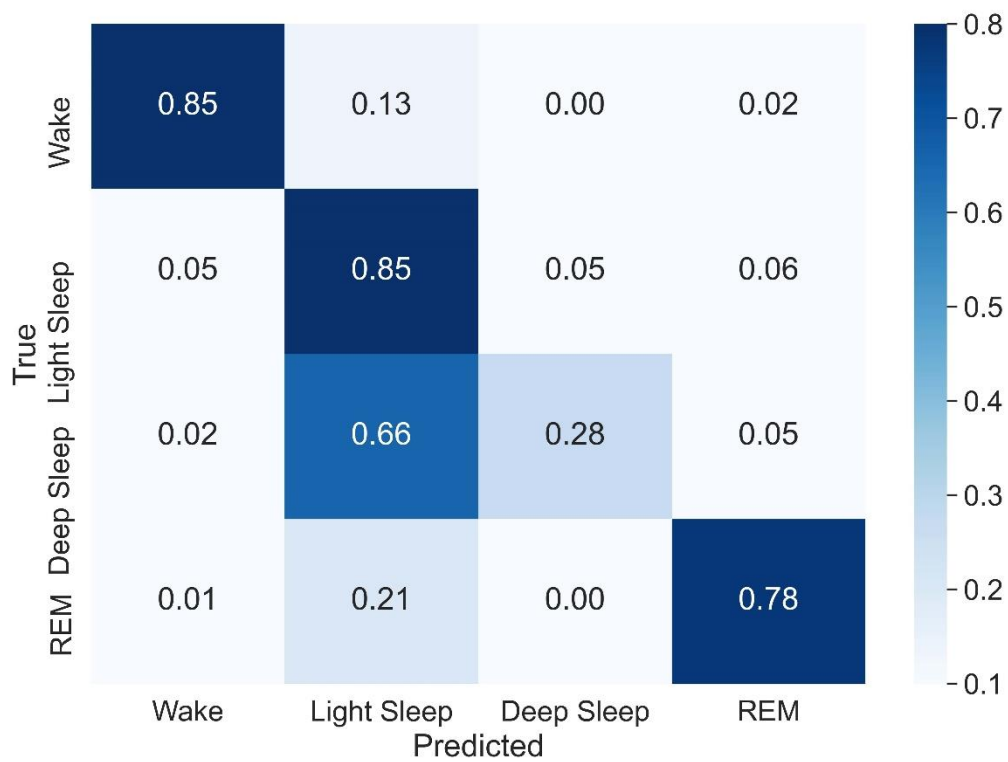


Figure 6

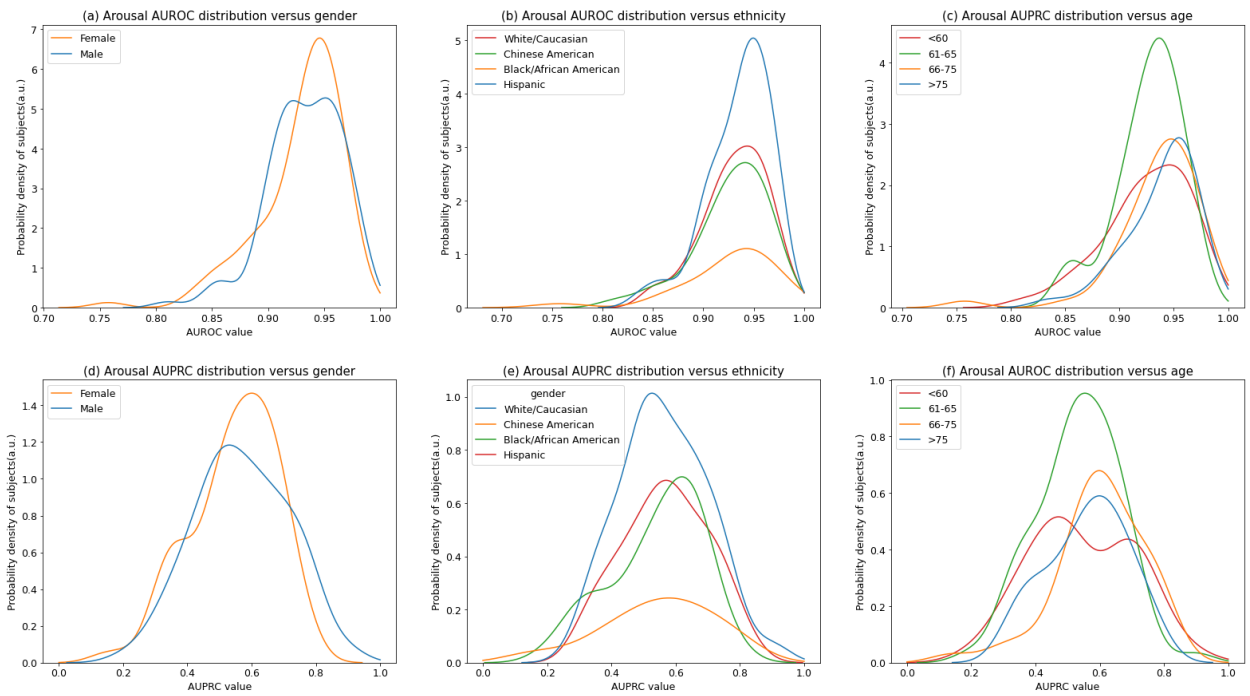


Figure 7

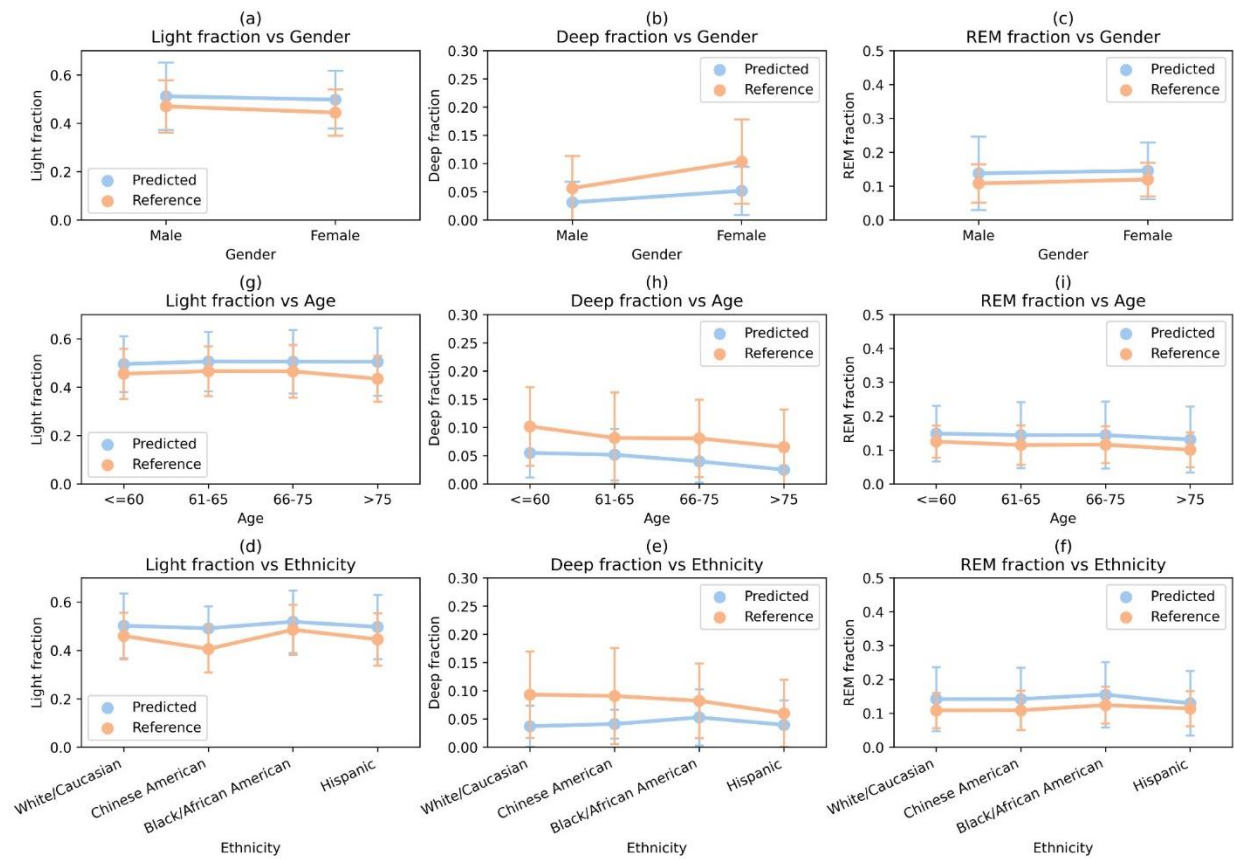


Figure 8

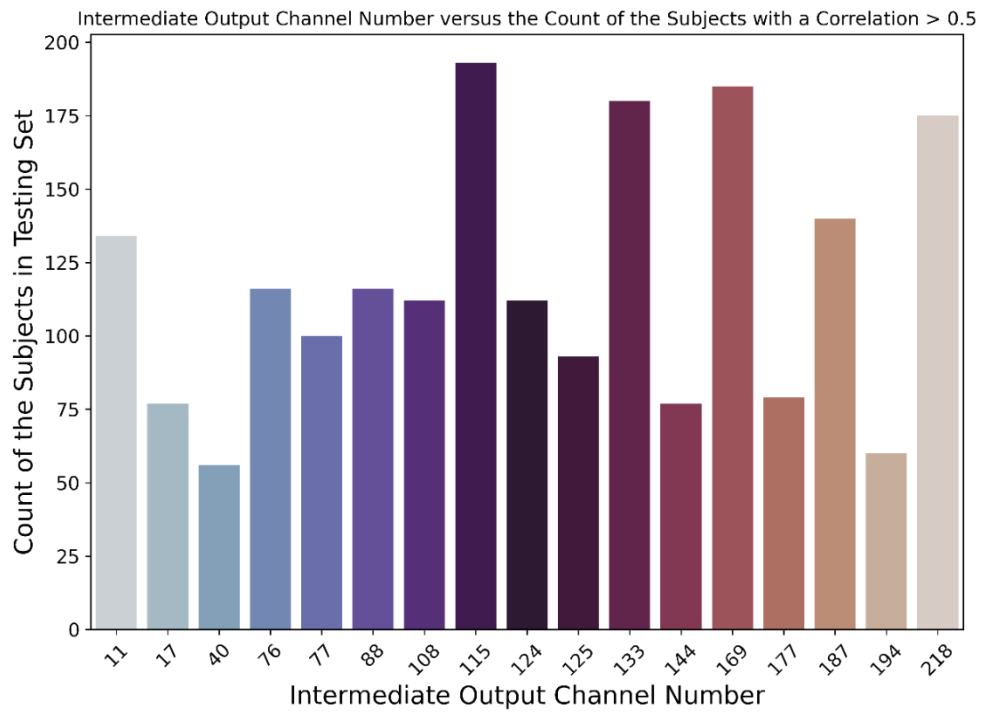


Figure 9

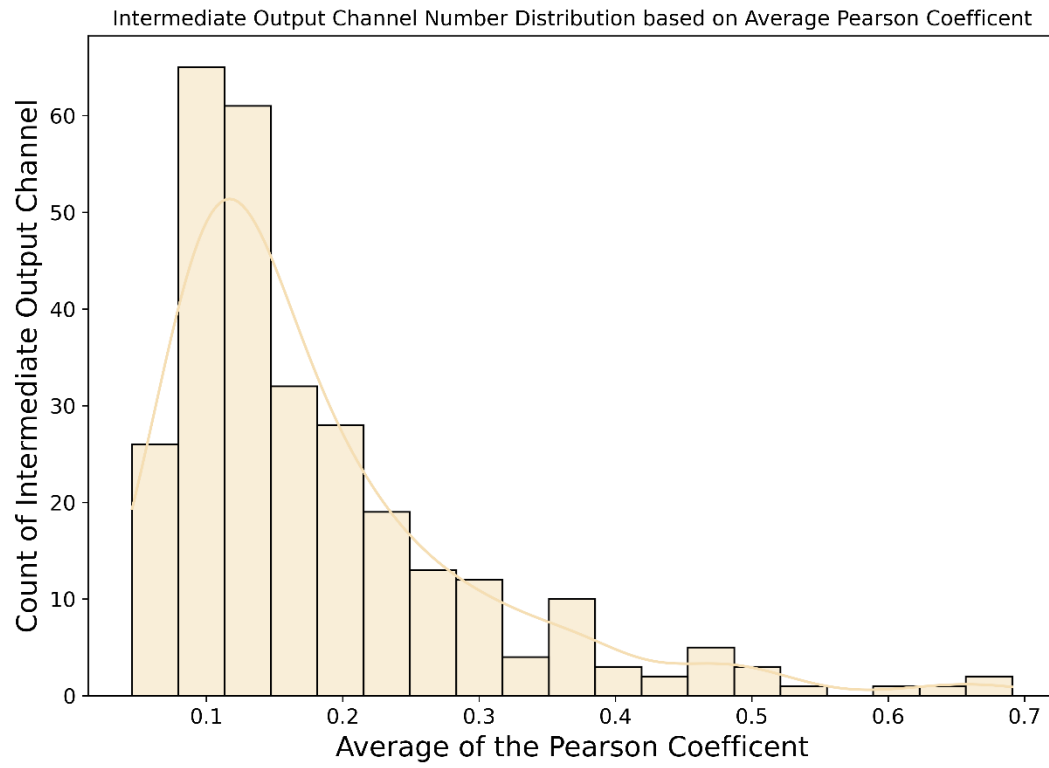


Figure 10

Example of intermediate output channel[133] and instantaneous heart rate of subject[117] on a segment of 200s

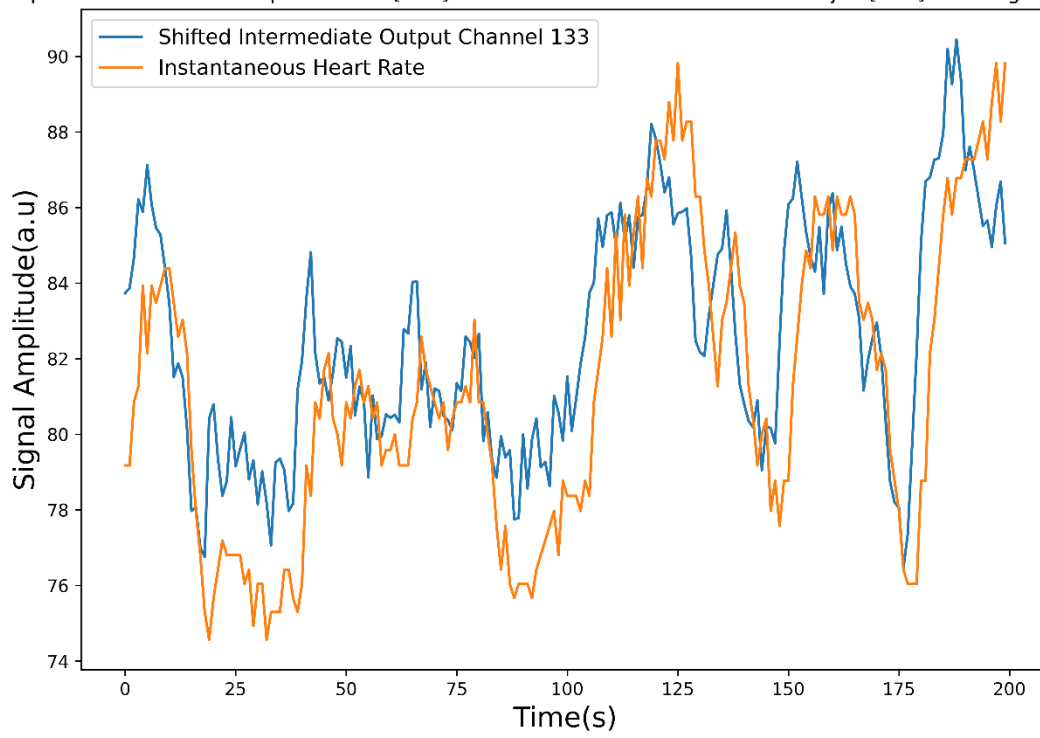


Figure 11

